

ECON 636 600: MACROECONOMIC THEORY I

Assignment 6 (due on 10:00am Tuesday, Nov 27)

1. (Qualifier 2010) (50 points) Trees in the utility function. Consider the following version of the Lucas Tree model. There are two kinds of trees. One kind is beautiful and yields utility. The other kind is ugly and yields no utility. Both trees yield an identical stream of identical fruit. The beautiful tree dividends at time t are d_{1t} , and ugly tree dividends at time t are d_{2t} . Further, $d_{1t} = d_{2t} = d_t/2$. Thus the physical output of the two trees are identical at each point in time.

The fruit itself is non-storable and generates utility. Trees last forever, and trees are the only source of fruit. There is no production in this economy, other than fruit from trees. There is one of each kind of tree for each individual in the economy. The initial allocation of tree is that each individual owns one beautiful and one ugly tree. There are N individuals in the economy. Assume the dividend process is such that $E_t[d_{t+j}] = d_t$, for any $j > 0$.

Consumers are identical, so we can work with a representative consumer. Preferences are given by

$$E_0 \sum_{t=0}^{\infty} \beta^t [\ln(c_t) + \gamma \ln(s_{1t})], \gamma > 0$$

Here c_t is consumption in period t and s_{1t} is shares of beautiful trees at the beginning of time t . Let p_{it} be the price of shares of a tree of type i during period t . Let R_{it} be the gross return on trees of type i held from t to $t+1$. Consider an equilibrium of this economy with markets in stocks of each type of tree.

- (a) Set up the household maximization problem as a dynamic programming problem. Derive the pricing function mapping the state of the economy at time t into prices p_{1t} and p_{2t} .

- (b) Show that $\gamma > 0$ is sufficient for $R_{1t} < R_{2t}$. That is, beautiful trees are dominated in rate of return. Also explain your result intuitively.

the DP problem is

$$\begin{aligned} V(s_{1t}, s_{2t}) = \max \quad & \ln(c_t) + \gamma \ln(s_{1t}) + \beta V(s_{1,t+1}, s_{2,t+1}) \\ \text{s.t.} \quad & c_t + s_{1,t+1}p_{1t} + s_{2,t+1}p_{2t} = (p_{1t} + d_t/2)s_{1t} + (p_{2t} + d_t/2)s_{2t}. \end{aligned}$$

the first order condition is

$$\begin{aligned} \frac{p_{1t}}{c_t} &= \beta E\left[\frac{p_{1t+1} + d_{t+1}/2}{c_{t+1}} + \gamma\right] \\ \frac{p_{2t}}{c_t} &= \beta E\left[\frac{p_{2t+1} + d_{t+1}/2}{c_{t+1}}\right] \end{aligned}$$

Using the equilibrium condition that $c_t = d_t$, we have

$$\begin{aligned} \frac{p_{1t}}{d_t} &= \beta E\left[\frac{p_{1t+1}}{d_{t+1}} + 1/2 + \gamma\right] \\ \frac{p_{2t}}{d_t} &= \beta E\left[\frac{p_{2t+1}}{d_{t+1}} + 1/2\right] \end{aligned}$$

it is easy to guess and verify that $\frac{p_{1t}}{d_t}$ and $\frac{p_{2t}}{d_t}$ are two constants (independent of time).

$$\begin{aligned} \frac{p_{1t}}{d_t} &= \beta \frac{1/2 + \gamma}{1 - \beta}, R_{1t} = E_t\left[\frac{p_{1t+1} + d_{t+1}/2}{p_{1t}}\right] = 1 + E_t\left[\frac{d_{t+1}/2}{p_{1t}}\right] \\ \frac{p_{2t}}{d_t} &= \beta \frac{1/2}{1 - \beta}, R_{2t} = E_t\left[\frac{p_{2t+1} + d_{t+1}/2}{p_{2t}}\right] = 1 + E_t\left[\frac{d_{t+1}/2}{p_{2t}}\right] \end{aligned}$$

the rate of return $R_{1t} < R_{2t}$ because $p_{1t} > p_{2t}$.

- Consider an economy with a continuum of identical consumers. The utility of a representative consumer is

$$E \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right],$$

where $\beta \in (0, 1)$, c_t is consumption and E is the expectation operator. Each consumer owns 2 trees, indexed by g and b . The fruit of tree $i \in \{g, b\}$, y_{it} , evolves as

$$y_{it} = \begin{cases} \theta_t, & i = b; \\ 1 - \theta_t, & i = g, \end{cases}$$

where θ_t is an aggregate shock whose value is either H or L . $0 < L < H < 1$. Assume that θ_t follows a two-state Markov chain, with

$$\Pr[\theta_{t+1} = H | \theta_t = H] = \pi, \quad \Pr[\theta_{t+1} = L | \theta_t = H] = 1 - \pi,$$

$$\Pr[\theta_{t+1} = H | \theta_t = L] = 1 - \pi, \quad \Pr[\theta_{t+1} = L | \theta_t = L] = \pi.$$

There is one share in each tree and these shares are competitively traded. Let p_{it} be the price of a tree $i \in \{g, b\}$ in period t . In addition to shares, there is also a riskless one-period bond. A bond sold in period t is a claim to one unit of consumption (i.e., fruit) in period $t + 1$. Let q_t be the price of a bond in period t .

- (a) Solve for p_{it} and q_t in each aggregate state. Also determine the real interest rate in each state.
- (b) Recall that the risk premium is the difference in rates of return between the risky asset and the riskless bond, i.e., $E_t[\frac{p_{it+1} + y_{it+1}}{p_{it}} - \frac{1}{q_t}]$. Determine the risk premium for each tree.

- (a) Since all consumers are identical, each consumer consumes the fruits of his own trees. $c_t = 1$ for all t . The riskless bond q_t satisfies

$$q_t u'(c_t) = \beta u'(c_{t+1}) 1,$$

hence $q_t = \beta$ and the real interest rate is $1/\beta$. Price p_{bt} satisfies

$$p_{bL} = \beta[\pi(p_{bL} + L) + (1 - \pi)(p_{bH} + H)],$$

$$p_{bH} = \beta[(1 - \pi)(p_{bL} + L) + \pi(p_{bH} + H)],$$

which implies

$$p_{bH} = \frac{\frac{\beta(\pi H + (1 - \pi)L)}{1 - \beta\pi} + \frac{\beta(1 - \pi)}{1 - \beta\pi} \frac{\beta(\pi L + (1 - \pi)H)}{1 - \beta\pi}}{1 - \left(\frac{\beta(1 - \pi)}{1 - \beta\pi}\right)^2},$$

$$p_{bL} = \frac{\frac{\beta(\pi L + (1 - \pi)H)}{1 - \beta\pi} + \frac{\beta(1 - \pi)}{1 - \beta\pi} \frac{\beta(\pi H + (1 - \pi)L)}{1 - \beta\pi}}{1 - \left(\frac{\beta(1 - \pi)}{1 - \beta\pi}\right)^2}.$$

It is easy to get $p_{gt} = \frac{\beta}{1 - \beta} - p_{bt}$.

(b) There is no risk premium for any share because consumption is constant.

$$p_{bt} = \beta E_t[(p_{bt+1} + \theta_{t+1})|\theta_t],$$

$$p_{gt} = \beta E_t[(p_{gt+1} + 1 - \theta_{t+1})|\theta_t],$$

which implies

$$E_t\left[\frac{(p_{bt+1} + \theta_{t+1})}{p_{bt}}|\theta_t\right] - \frac{1}{\beta} = 0,$$

$$E_t\left[\frac{(p_{gt+1} + 1 - \theta_{t+1})}{p_{gt}}|\theta_t\right] - \frac{1}{\beta} = 0.$$